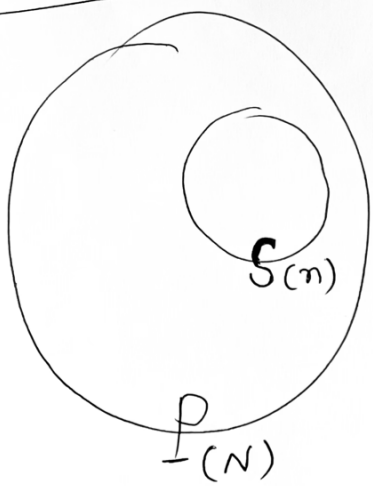
CLT



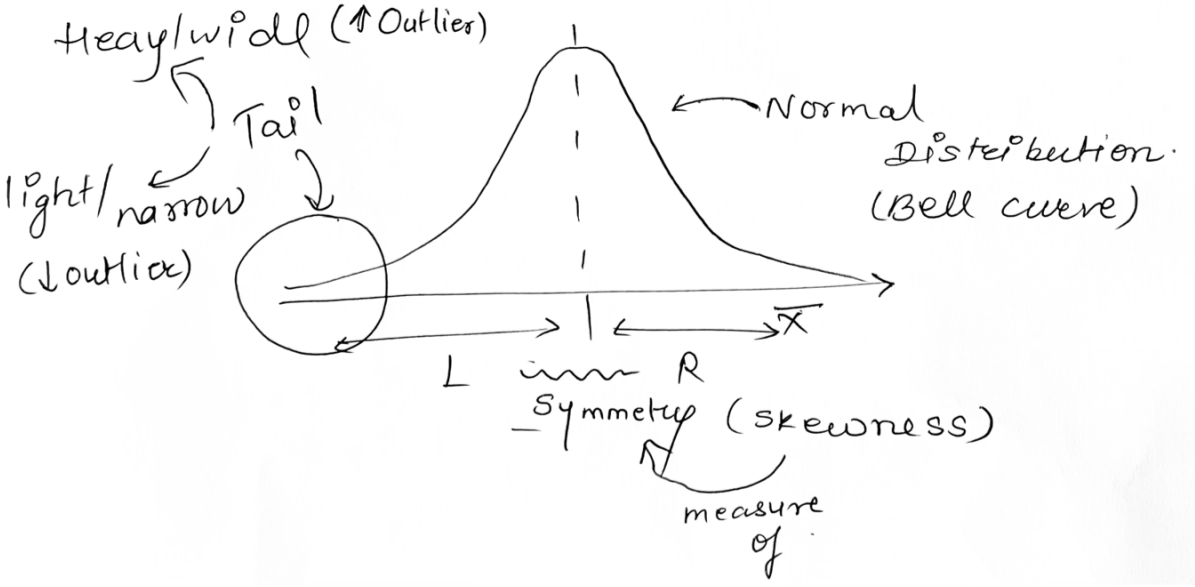
Sample size
$(n) \geq 30$

$S_1 = \{ \quad \} \Rightarrow \overline{x_1}$

$S_2 = \{ \quad \} \Rightarrow \overline{x_2}$

$\vdots$

$S_m = \{ \quad \} \Rightarrow \overline{x_m}$



- Inverse of Matrix

$$A^{-1} = \frac{Adj(A)}{|A|}$$

Adj(A) → Adjoint of A.

|A| → Determinant of A

* Adjoint of A
  - Minor
  - Cofactor Matrix
  - Transpose.

Eg:- $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 2 & 9 \end{pmatrix}$

$$|A| = (1)[45 - 12] - (2)[36 - 42] + (3)[8 - 35]$$

$$= \underline{-36}$$

$M_{1,1} = \det \begin{vmatrix} 5 & 6 \\ 2 & 9 \end{vmatrix} = 33$

$M_{1,2} = \det \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = -6$

$M_{1,3} = \det \begin{vmatrix} 4 & 5 \\ 7 & 2 \end{vmatrix} = -27$

$M_{2,1} = \det \begin{vmatrix} 2 & 3 \\ 2 & 9 \end{vmatrix} = 12$

$M_{2,2} = \det \begin{vmatrix} 1 & 3 \\ 7 & 9 \end{vmatrix} = -12$

$M_{2,3} = \det \begin{vmatrix} 1 & 2 \\ 7 & 2 \end{vmatrix} = -12$

$M_{3,1} = \det \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} = -3$

$M_{3,2} = \det \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} = -6$

$M_{3,3} = \det \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = -3$

Cofactor :- $(-1)^{i+j} M_{ij}$

$$M_{1,1} = (-1)^{1+1} \times 33 = \underline{33}$$

(i ↗ , ↖ j)

$$C' = \begin{pmatrix} 33 & 6 & -27 \\ -12 & -12 & 12 \\ -3 & 6 & -3 \end{pmatrix}$$

Transpose ↓

$$\begin{pmatrix} 33 & -12 & -3 \\ 6 & -12 & 6 \\ -27 & 12 & -3 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{1}{-36} \begin{pmatrix} 33 & -12 & -3 \\ 6 & -12 & 6 \\ -27 & 12 & -3 \end{pmatrix}$$